Compound Structural-Semantic Fingerprinting for On-Device AI Copyright Verification*

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Abstract

On-device AI deployment has heightened concerns about model intellectual property, where adversaries may extract a deployed model and reuse, fine-tune, or obfuscate it for commercial purposes. Conventional graph-based fingerprinting captures topological structure but is known to misclassify independently trained models that share the same architecture. We propose a compound fingerprint that fuses a structural score (PKG hash and approximate graph edit distance) with a semantic score derived from weight distribution, quantization parameters, and tensor-shape statistics, all extracted directly from TFLite buffers. The structural score serves as a fast first-stage filter, and the semantic vector enables refined verdicts. Evaluation on MobileNetV1 quantized versus floating-point pairs shows high structural overlap but clearly separated semantic signatures, demonstrating that structural-semantic fusion reduces the false positives caused by same-architecture ambiguity. Additional architecture-diverse screening on ResNetV2-101 and MobileBERT XS QAT Lite further rejected comparison models at the fast structural-filter stage, with PKG similarities of 0.2027 and 0.1360 and screening-stage fused scores of 0.5928 and 0.4550, respectively.

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1 Introduction

On-device AI inference proliferates on mobile and embedded platforms, exposing model assets to extraction and unauthorized reuse and motivating model fingerprinting. Graph-based fingerprinting offers a compact signature resilient to pruning, fine-tuning, and light-weight obfuscation, yet a persistent failure mode remains in which independently trained same-architecture models produce nearly identical operator graphs and undermine real-world attribution.

We address this same-architecture ambiguity for TFLite-deployed models by augmenting a fast structural score with a semantic score derived from quantized weight distributions, deployment-time quantization parameters, and tensor-shape statistics. Our method combines PKG-based filtering for efficient first-stage screening with semantic refinement for final discrimination, while the semantic vector is normalized to remain comparable across different architectures.

Keywords: Model fingerprinting, Intellectual property protection, On-device AI, Deep learning security

2 Proposed Framework

The framework operates on TFLite buffers in four stages as shown in Fig. 1. First, a graph parser constructs an operator graph containing both topological information and per-node attributes. In the structural path, a Weisfeiler–Lehman kernel hash, PKG, is computed for first-stage filtering, followed by approximate graph edit distance, GED, for refinement. These two components are combined as follows: $S_{struct} = 0.7S_{pkg} + 0.3S_{ged}$

In the semantic path, three feature families are extracted for each tensor. The first is a weight descriptor consisting of mean, standard deviation, skewness, kurtosis, percentiles, and a 32-bin histogram. The second is a quantization summary including per-tensor and per-channel scale and zero-point, along with quant-mode flags. The third is a tensor-shape descriptor using rank, dimension ratios, and log-parameters.

Robust scaling is first applied for each operator type, after which all features are concatenated, and the final vector is L2-normalized. The semantic score is then computed as a weighted combination, giving greater importance to the weight-distribution channel. The weight-distribution descriptor receives the highest weight as the primary discriminative factor for same-architecture pairs, while quantization and shape contribute as secondary and architecture-gating signals, respectively.

$$S_{sem} = 0.60S_{weight} + 0.25S_{quant} + 0.15S_{shape}$$

The two paths are finally fused as follows: $S_{total} = 0.45S_{struct} + 0.55S_{sem}$

Here, the semantic channel is given slightly more weight than the structural channel, reflecting its role as the principal disambiguator of same-architecture false positives while the structural channel acts as the first-stage filter and the carrier of obfuscation resilience.

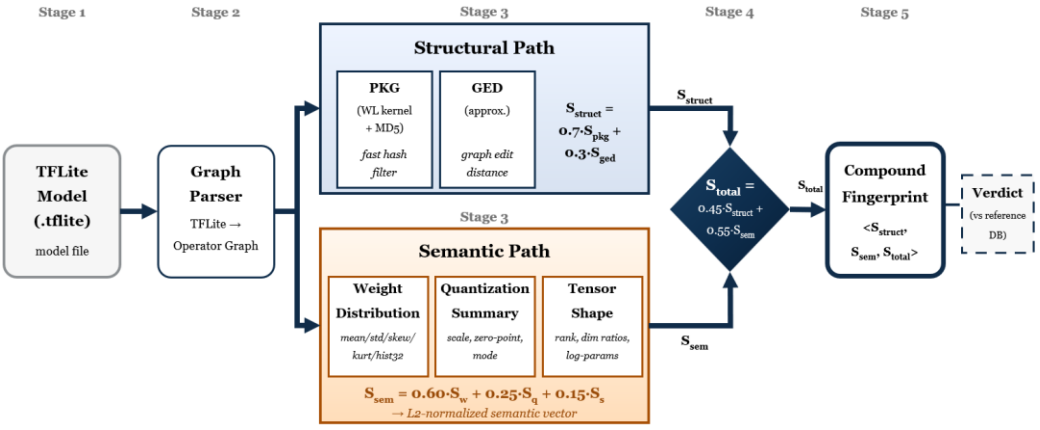


Figure 1. Compound Structural-Semantic Fingerprinting Pipeline

3 Architecture-Diverse Screening

To address generalization beyond the MobileNetV1 pilot, we additionally evaluated architecture-diverse comparison models using ResNetV2-101 and MobileBERT XS QAT Lite TFLite models. Against the MobileNetV1 reference, the fast PKG structural filter produced low PKG similarity scores, denoted S_{pkg}, of 0.2027 for ResNetV2-101 and 0.1360 for MobileBERT XS QAT Lite. Because both pairs were rejected at the PKG screening stage, the computationally expensive GED refinement was not invoked. We therefore report a pre-GED structural-semantic screening score, $S_{pre\text{-}GED} = 0.45*S_{pkg} + 0.55*S_{sem}$, where S_{sem} is the weighted semantic score computed from S_{weight}, S_{quant}, and S_{shape}, corresponding to weight-distribution, quantization, and tensor-shape similarities, respectively. The resulting $S_{pre\text{-}GED}$ values were 0.5928 for ResNetV2-101 and 0.4550 for MobileBERT XS QAT Lite, both below the provisional reuse threshold and consistent with non-reuse decisions across residual CNN and Transformer/BERT-style architectures.

4 Conclusion

We presented a compound structural-semantic fingerprinting pipeline for AI copyright verification on TFLite-deployed on-device models. Evaluation of MobileNetV1 quantized against floating-point pairs shows $S_{total} = 0.6372$, with $S_{struct} = 0.84$ reflecting near-identical topology while the semantic channel separated the two via $S_{weight} = 0.5355$ and a quantization-channel mismatch expected by design between quantized and floating-point operands. These results confirm that the semantic channel resolves same-architecture pairs that structural-only methods would conflate. Additional screening on ResNetV2-101 and MobileBERT XS QAT Lite shows that the proposed fast structural filter rejects architecture-diverse comparison models at the PKG screening stage before GED refinement.